



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

**LABTRACK: PROPERTY INVENTORY AND MONITORING SYSTEM
USING QUICK RESPONSE (QR) CODE**

A Capstone Project Presented to the
Faculty of the Information and Communication Technology
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Justiniana Edera, San Jose, Dinagat Islands

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Bachelor of Science in Information Technology

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The Researchers



ABSTRACT

**LABTRACK: PROPERTY INVENTORY AND MONITORING SYSTEM
USING QUICK RESPONSE (QR) CODE**

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This research addressed challenges encountered at Don Jose Ecleo Memorial College regarding laboratory equipment borrowing procedures. The existing system was insecure, resulting in equipment misplacement and the loss of records. The proposed solution, 'LabTrack: Property Inventory and Monitoring System Using QR-Code,' aimed to improve the process by efficiently monitoring equipment and borrowers.

It maintained a secure record of all transactions and utilized QR codes on each piece of equipment for easy access to borrower details and equipment status. This enabled the property custodian and lab in-charge to effectively monitor transactions and maintain accurate borrower records.

The implementation of LabTrack was anticipated to enhance the security and manageability of laboratory operations.

Keywords: *inventory, monitoring system, equipment, QR code, laboratory, offline web.*



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CHAPTER 1

INTRODUCTION

In today's generation, modern universities adopted technology to enhance campus equipment borrowing processes. The implementation of laboratory monitoring systems was advocated in all schools for secure and manageable equipment inventory and borrower tracking.

Researchers at Don Jose Ecleo Memorial College identified security issues in laboratory equipment borrowing, which led to equipment loss and record mismanagement. To address this, they proposed "LabTrack," a QR-Code based inventory system, aimed at enhancing security and efficiency in all campus laboratories.

The property inventory and monitoring system, as described by (Jr., 2023), was designed to improve and streamline processes in every establishment. This system facilitated the monitoring of equipment in each laboratory and tracked the borrowers. Furthermore, its enhanced equipment tracking by maintaining secure records, which included information about the borrowing and return data for each piece of equipment

LabTrack, utilizing QR codes affixed to each equipment piece, streamlined the borrowing process by displaying borrower and equipment details. It enabled custodians to monitor laboratory transactions and track equipment circulation. Mandatory borrower registration further aided in tracking.



1.1 Project Context

The LabTrack was designed to enhance the borrowing process in all laboratories at Don Jose Ecleo Memorial College by utilizing QR codes for borrowing tools and equipment. This strategy was expected to make the process faster, more accountable, and more efficient. In the current technological landscape, such advancements were highly recommended. QR codes, due to their ease of use and versatility, could simplify the process of equipment borrowing. The project required every borrower to register an account and log in when borrowing, thereby creating a traceable record of the borrowing activity. This system assisted laboratory in-charges in maintaining an accurate record of each borrower and the borrowed equipment, and aided the property custodian in monitoring these items. Ultimately, this project provided a robust and efficient mechanism for monitoring tools and equipment, preparing for any future contingencies, and enhancing operational efficiency in the laboratories at Don Jose Ecleo Memorial College.

1.2 Purpose and Description

The study aimed to create a QR code-based system, LabTrack, to streamline equipment borrowing in all labs at Don Jose Ecleo Memorial College. LabTrack centralized the borrowing process, offering features such as:

- **Borrowing Equipment:** Users could borrow equipment by scanning its QR code with a mobile phone.
- **User Access Control:** The system provided appropriate access levels to borrowers, property custodians, and lab in-charges.



- **Inventory Reporting:** LabTrack generated comprehensive equipment information, accessible by the lab in-charge and the property custodian.
- **User-Friendly Interface:** The system offered an intuitive interface for all users.
- **Security Measures:** LabTrack ensured data protection and confidentiality in compliance with regulations and standards.

1.3 Research Objectives

The development of the LabTrack on every laboratory, in Don Jose Ecleo Memorial College is guided by a set of general and specific objectives.

1.3.1 General Objectives

The researcher's general objectives were to develop a web platform for property inventory and monitoring for physical laboratories, improving manual tracking, inventory management, and operational efficiency using a Quick Response (QR) code.

1.3.2 Specific Objectives

This project primarily aims to develop a system called "LabTrack: Property Inventory and Monitoring System using QR-code." The study seeks to;

- Track every laboratory's equipment.
- Develop an inventory module for monitoring borrower and equipment records.
- Implement a QR-code scanner in a local web system.
- Ensure the system meets ISO 25000 standards for functional requirements.



1.4 Scope and Limitation of the Study

1.4.1 Scope

The system LabTrack was designed to manage and monitor borrowers, laboratories, and equipment. Key features include:

- **Tracking and Monitoring:** The system tracked borrower details and equipment in each laboratory. It allowed the Property Custodian to view borrowing history and manage their account.
- **Equipment Management:** Laboratory In-charges could add, delete, and edit equipment records, manage returns, and note any damage or loss. The Property Custodian oversaw this process.
- **Inventory Function:** Both the Laboratory In-charge and the Property Custodian could monitor equipment availability and condition. This information was stored as an inventory.
- **QR-Code Scanner:** The system included a QR-code scanner for borrowing and returning equipment.
- **User Security:** A login module was provided for all users, with a registration function for new borrowers. The Property Custodian created accounts for each Laboratory In-charge and recorded their designated laboratories.

1.4.2 Limitation

The system has the following limitations:

- **Dependence on Laboratory In-charge:** The borrowing process could have been ineffective if the Laboratory In-charge had been absent, potentially leading to equipment loss.



- **Data Loss Risk:** The system has no auto backup or scheduled backup feature, which may result in potential data loss if the database is not manually backed up.
- **RFID Tracking:** The system lacks the capability to real-time track the location of equipment.

1.5 Significance of the Study

The proposed system was a local network website developed with modern technologies, offering significant benefits to various stakeholders.

Don Jose Ecleo Memorial College - The system aided in maintaining tool availability and monitoring borrowing in laboratories.

Property Custodian - The system enabled easy borrower monitoring and assisted in preventing equipment loss.

Laboratory In-Charge - The system allowed direct monitoring of borrowers and equipment, facilitating record-keeping.

Borrower - The system simplified the borrowing process via a QR code scanner.

Researcher - The system enriched learning abilities and improved technical capabilities.

Future Researchers - The study served as a reference for similar future studies.



CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Review of Related literature in Objective 1

According to the study by (Ntafatiro, 2023) a tracking system using RFID was developed titled IoT-Based Computer Laboratory Equipment Tracking System to ensure and enhance the safety of every tool and equipment inside the computer Laboratory. They created this method to improve the borrowing process, which was previously manual, and to ensure the security of all tools and equipment inside the computer Laboratory. In this regard, the system study implemented the Advanced Encryption Standard and encryption and decryption algorithm using a 128-bit key to provide the computer Laboratory with a better solution for securing the transmission of RFID data. RFID was used to notify every borrower who was allowed to borrow tools and equipment inside the computer Laboratory.

As stated in the study by (Zainol, Al-Amoodi, Bahrin, Arman Alim, & Ibrahim, 2021) they implemented a global positioning framework to provide safety for each lab hardware. This framework was intended to help increase the security and management system concerning acquiring. Thus, the step-by-step process of acquiring was described.

Borrowers used an Android phone and scanned the RFID to access lab hardware, which helped monitor hardware and track borrowers in their precise location. The system required first obtaining the information of each borrower prior to acquiring.

Additionally, according to (Sulistyo, Achmad, & Purnama, 2022), a system for coordinating asset management and tracking was implemented at Politeknik Negeri



Balikpapan to enhance proper management and monitor the history of assets, locations, and users. The Quick Response code helped the TVET by making it easier to track the history of assets, locations, and users.

A new, innovative method was implemented to assist in keeping track of who accesses the laboratory and rooms, based on research conducted by (Fauzi, Mohamed, Hashim, & Saleh, 2020). They were able to monitor the incoming and outgoing actions of users who used a web-based website server thanks to a logging system. It was recommended that everyone planning to use a room or laboratory scan the QR code scanner located there. Through the use of a phone integrated into the system, they were able to track and monitor room users when they scanned into their respective locations.

Finally, according to the study by (Byers, Cheriya-mulla, Ewason, Hall, & Wolf, 2022), tracking building component information throughout its lifecycle is crucial for reducing reuse risks. While existing literature primarily focused on RFID for equipment tracking, this study employed QR codes due to their cost-effectiveness, speed, and accessibility. QR codes were laser-engraved onto structural wooden members. This proof-of-concept evaluated the benefits and challenges of using QR codes as a preferred tracking technology.

2.1 Review of Related literature in Objective 2

According to the findings of the analysis conducted by (Ahmadi, 2018), this study was conducted to provide each user with the most up-to-date inventory management for surgical supplies and equipment on a regular basis. With the help of this study, they were able to identify the systematic manner in which each user who used a surgical facility went about things and the papers that each user may have found useful in terms of cost savings.



To improve laboratory equipment inventory, in addition to the study by (Wicaksono, Mochamad, Syahrul, Rahmatya, & Myrna, 2021) the "development of laboratory equipment inventory system using RFID and Internet of things" was put into practice. The users of this system received notifications regarding the data of borrowers, the length of the loan, the due date, and the availability of tools and equipment in each laboratory. Maintaining the inventory availability of tools, equipment, and their borrowers was the primary objective of this system research.

In the study by (Rabiah, Lindawati, & Sarjana, 2022) a web-based laboratory inventory application was designed to enhance the long-term process of borrowing tools and equipment. They implemented this system to help control the inventory and the process of borrowing tools and equipment within the laboratory office. This web-based lab inventory application helped every student in the engineering study program lab maintain the quick borrowing process.

An inventory management system (IMS) developed by (Singh, Kunder, Jain, & Sagvekar, 2023), for the University of Mumbai (UoM) aimed to make the inventory and borrowing process more efficient. The school struggled to maintain and protect all the vital documents on the campus. Therefore, they introduced a system to enhance the inventory and borrowing process. The system reduced the time required for creating an inventory. Moreover, the system ensured that the records of all the essential documents were preserved well.

Based on the research conducted by (Kurnaedi, Oktora, Nasrullah, Prayoga, & Pramanto, 2023), there is another study that focuses on inventory management using an application that monitors, tracks, borrows, and purchases goods. This application can display the data of goods that are received and delivered, as well as the inventory



system's history. By using this application, the inventory asset can enhance the effectiveness of inventory management. It can also demonstrate the significance of keeping an eye on what is going on in the inventory.

2.3 Review of Related literature in Objective 3

According to (Ng, 2021), a student attendance system based on QR codes with unique identification was developed, utilizing mobile devices to scan QR codes to achieve attendance in class or any event. The QR code system was a combination of android application development for taking and storing attendance in the cloud storage. The reason this advanced method was chosen as a tracking system was because QR code-based systems did not require high implementation on hardware.

The research project by (Madzidon & Harun, 2022) claimed that borrowing practices had advanced technologically. The borrowing procedure used in this study was expected to be easier to assist and more beneficial than the manual approach. According to the survey, Sk Jelotong students still checked out books by hand. According to the survey, Sk Jelotong students still checked out books by hand.

Therefore, it was decided to make borrowing easier by utilizing QR codes in addition to expanding the availability of books. If QR codes were utilized, they would help collect, store, and make information usable for customers, specifically the administrators and librarians of S.K. Jelotong, requiring both a username and password for authentication.

In addition to the study by (Mishra, Vaishnavi, Bakale, H., & Divya, 2020), a QR code-based student monitoring system was implemented to make checking



attendance easier. It also provided a way to check all quizzes and assignments. This would help parents, as they would receive a notification through SMS if their child was at school, enabling them to monitor their children's attendance and academic activities more effectively.

The purpose of this study by (Din, Anwar, & Fazal, 2021) was to improve the book borrowing process for library users. Asset tagging for libraries was utilized, making it faster and easier for users to borrow or reserve books. By scanning a QR code, users could access information about any book in the library, such as the call number and title, facilitating easier location of the book.

Based on the study by (Ahmed & Zanelidin, 2019), a system study was designed to enhance daily life activities and facilitate time consumption for individuals in a rush. Mobile phones were used to scan QR codes attached to every item. When scanning the QR code tag, it appeared in the network system of every mobile phone, receiving data and displaying details of every item on the smartphone's screen. Users could also add items to their list on their mobile phones.

2.4 Synthesis

The inventory management system tracked and monitored the equipment using QR codes. It aimed to enhance the process and supply of tools and equipment. It also helped to track and monitor equipment. The web-based laboratory inventory application helped control the inventory and the process of borrowing tools within the laboratory office.



CHAPTER 3

TECHNICAL BACKGROUND

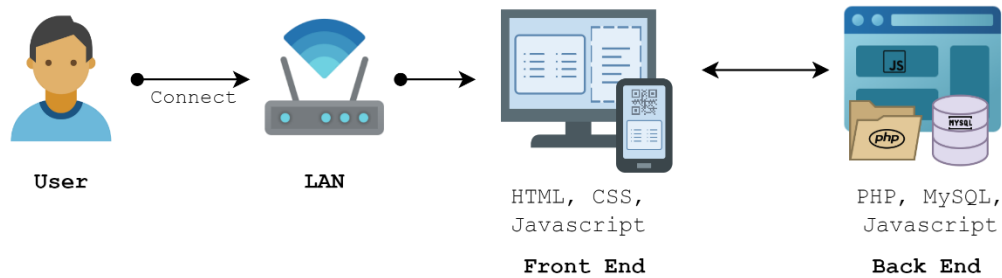


Figure 1. System Architecture of LabTrack

LabTrack was constructed on a robust technical foundation, utilizing JavaScript and CSS for UI/UX design. The goal was to create a user-friendly experience for property custodians, lab in-charges, and borrowers. Additionally, PHP and SQL were employed to establish powerful and efficient database connections and interactions.

PHP (Hypertext Preprocessor): PHP functioned as the server-side scripting language for this system. It enabled the creation of dynamic content, oversaw user authentication and authorization, especially in logging in, scanning QR codes, and adding/modifying borrower and equipment data. Additionally, it interfaced with the MySQL database to access and modify data. Due to its flexibility and comprehensive function library, PHP was the preferred choice for the system's back-end, ensuring smooth borrowing transaction processing and management of borrower and equipment data.

JavaScript: From the perspective of the property custodian and lab in-charge, JavaScript was utilized to improve the user interface and develop interactive elements that augmented the borrower's experience, especially in monitoring equipment and



borrowers. This included libraries such as “toastr.min.js” and “jquery-1.11.1.min.js”. It was applied to verify inputs from the property custodian, lab in-charge, and borrower, and to deliver instantaneous updates on equipment and borrowers. Additionally, “html5qrcode.min.js” was used to facilitate features such as the QR code scanner, an interactive borrowing procedure, and interactive tables. JavaScript ensured a dynamic and captivating web interface for users interacting with the system.

CSS (Cascading Style Sheets): CSS was employed for the front-end development of the system, responsible for styling and structuring the web pages, including the login page, borrower’s page, property custodian/lab in-charge page, and especially the inventory’s page. The system utilized the “bootstrap.bundle.min.css” library as it ensured a plain, uniform, and visually pleasing design, simplifying navigation and order placement for users. Additionally, CSS facilitated responsive design, especially on the borrower’s page, as it was accessed by mobile devices. This ensured the system’s accessibility and user-friendliness across a variety of devices and screen sizes.

SQL (Structured Query Language): was the preferred language for querying the LabTrack system’s database management system. It facilitated communication with the database, allowing it to store and manage crucial data. This included details about equipment and borrowers, user profiles, records of borrowing and returning, and inventory specifics. Additionally, Apache was used to deploy the database onto a server for the LAN website. Due to its reliability, scalability, and robust support for intricate queries, SQL was well-suited for effectively managing substantial volumes of borrowing transaction data.

CHAPTER 4

METHODOLOGY

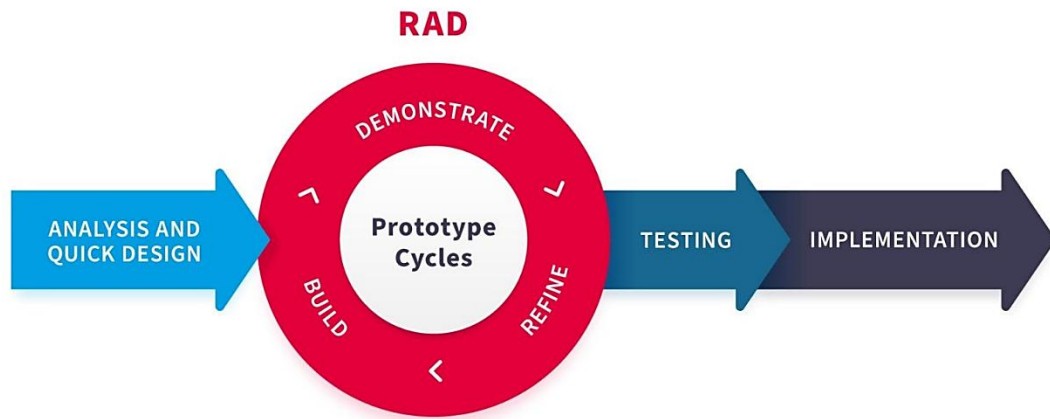


Figure 2. Rapid Application Development Diagram

The Rapid Application Development methodology was chosen for its emphasis on rapid prototyping and iteration, which minimized the planning phase and maximized the prototype development. This approach was suitable for the development of 'LabTrack: Property Inventory and Monitoring System using QR-code'. The RAD methodology was divided into four phases: System Analysis and Design, Development of the System, and System Implementation.

4.1 Phase 1: System Analysis and Design

In the Requirements Planning phase, the researchers conducted a series of interviews and discussions, along with formalizing letters to the beneficiaries for authorization, in which the researchers determined the goals and expectations of the project. The researchers also identified the current and potential issues that needed to be addressed through brainstorming and a series of discussions with the adviser, focusing on objectives 1 and 2, since there was a format to be followed in the borrowing



processes. The output of this phase was a requirement planning document that outlined the scope and objectives of the project.

4.2 Development of the System

During the User Design phase, prototypes were developed based on the requirements gathered in the previous phase. The researcher's adviser and the beneficiary were involved in this phase to ensure their needs were met at every step of the design process. To ensure that objective 3 was met, various libraries and source codes were included in the system so that borrowers could properly interact with it. Additionally, to ensure the smooth development of the system, especially in the borrowing process and monitoring equipment and borrowers, the researcher used a computer unit with a minimum of 8GB of RAM, a minimum of 3.00GHz CPU speed, and a 500GB hard drive. This hardware configuration ensured sufficient resources for the smooth execution of the development tasks and facilitated the effective implementation of the system. The output of this phase was a prototype that met the user requirements.

4.3 Phase 3: System Testing

The Rapid Construction phase involved the actual coding and building of LabTrack. The goal of this phase was to produce a system that could be tested by the users of the system: Laboratory In-Charge, Property Custodian, and borrower. It was then evaluated for further enhancements. A series of documentations were made during the process. The output of this phase was a working model of the system.



4.4 Phase 4: System Implementation

The Cutover phase represented the final stage, during which the system underwent evaluation through a series of surveys conducted among the beneficiaries, namely the property custodian and lab in-charge, to determine if the system met their standards, as they were the primary users. Any necessary adjustments were made during this evaluation process. The output of this phase was the final system, ready for deployment.

4.5 Phase 5: System Evaluation

To achieve objective 4, the researchers followed the ISO/IEC 25000 requirements to thoroughly examine the LabTrack: Property Inventory and Monitoring System Using QR Code. Special attention was given to functionality, real-time efficiency, user interface usability, dependability, maintainability, portability, compatibility, and security. By applying a structured tool aligned with ISO/IEC 25000, the evaluation verified that the system met the required performance and beneficiary standards. The assessment instrument involved recording the system functions and identifying performance characteristics.

The system underwent evaluation by thirty respondents, comprising the property custodian, lab in-charge, and borrowers. To ensure diverse representation, the respondents were selected through random sampling. They were provided with a validation survey instrument to assess and rank the system's performance based on the specified criteria.



Likert-Scale Description	Scale	Interval
Strongly Disagree	1	1.0 – 1.7
Disagree	2	1.8 – 2.5
Neutral	3	2.6 – 3.3
Agree	4	3.4 – 4.1
Strongly Agree	5	4.2 – 5.0

Table 1: Likert-Scale Description Utilized in System Evaluation

4.5.1 Development and Testing

The proposed “LabTrack: Property Inventory and Monitoring System using QR-code” underwent designing, development, and evaluation, all conducted using the ISO/IEC 25000 Software Quality Model.

Respondents	N	Percentage
Property Custodian	1	3.3333%
Laboratory In-Charge	6	20%
Borrowers	23	76.6666%
Total	30	100%

Table 2. Distribution of Respondents

By utilizing a validated assessment instrument and involving a diverse group of respondents, the research aimed to gather comprehensive feedback on the system's functionality, efficiency, usability, dependability, maintainability, portability, compatibility, and security. The goal was to gain a deep understanding of the system's operation, facilitating a thorough evaluation and identification of any areas that could benefit from further refinement or development.



CHAPTER 5

RESULTS AND DISCUSSION

5.1 Implementation of Results

This section presented an overview of the outcomes and achievements of the implementation of the “LabTrack: Property Inventory and Monitoring System Using QR Code.” The objective was to assess the effectiveness, efficiency, and overall success of the implementation process. It emphasized the key results obtained during the deployment of the system. Please refer to the Appendices for samples of the system's input and output, as well as its user interface.

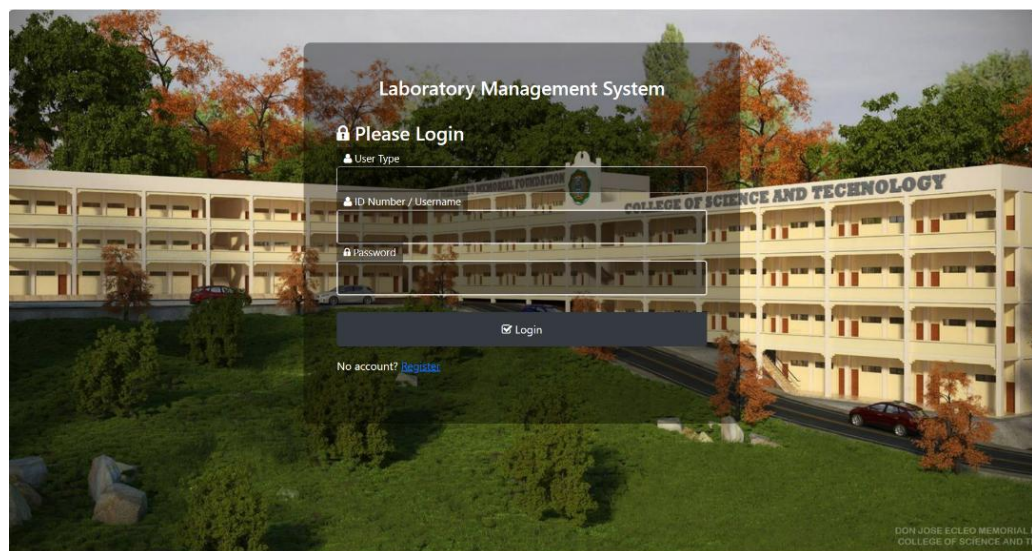


Figure 3. System Authentication (Desktop)



The page shown in Figure 3 was easy to navigate, and the login form was straightforward to use. User Type was designated for the property custodian/lab in-charge and borrower. Borrower registration occurred on this page.

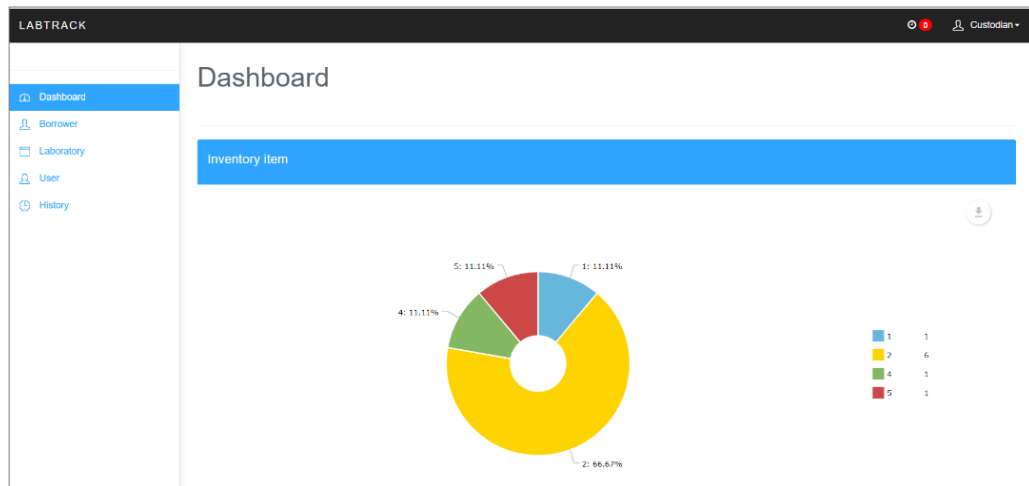


Figure 4. User Interface

The UI layout illustrated in Figure 4 displayed the dashboard, featuring a pie graph illustrating the equipment. This page was solely for viewing and served as the initial page displayed upon logging in as a property custodian or laboratory in-charge.

The screenshot shows the LABTRACK Borrower Details page. It includes a sidebar with navigation links: Dashboard, Borrower, Laboratory, User, and History. The main content area has a 'Borrower' header and a table with columns: ID Number, Name, Department, Type, and Year/Section. There are also buttons for 'Copy', 'CSV', 'Excel', 'PDF', and 'Print'. A search bar is present. Below the table, it says 'Showing 1 to 3 of 3 entries'. On the right, there is an 'Action' dropdown menu with options: Edit, deactivate, and Borrower Profile. Navigation buttons 'Previous', '1', and 'Next' are at the bottom right.

ID Number	Name	Department	Type	Year/Section
0000	pedro penduko	BEED	Borrower	
1234	Juan dela cruz	BSHRM	Borrower	
2002	junmar but ay	BSIT	Borrower	

Figure 5. Borrower Details



The interface depicted in Figure 5 displayed the borrowers registered in the system and allowed for reviewing the equipment they had borrowed or returned. This feature was accessible to both the property custodian and the laboratory in-charge.

Figure 6. Room Details - Property Custodian

The screenshot shows the 'LABTRACK' application interface. On the left is a sidebar with navigation links: Dashboard, Borrower, Laboratory, User, and History. The main content area is titled 'Laboratory' and includes a search bar and a table of laboratory details. The table has columns for Laboratory Name, Laboratory Incharge, Time Limit, and Action. There are four rows of data. An 'Add Laboratory' button is in the top right, and a dropdown menu is open for the first row's 'Action' column.

Laboratory Name	Laboratory Incharge	Time Limit	Action
Computer Laboratory 1	Judylye Novo	1 days	Action + Edit View Items List Borrow
Laboratory 2	Carl Dawson	1 days	
Laboratory 3	Escanor Fauzi	2 days	
Lab 4	Kallo	3 days	

The interface shown in Figure 6 played a vital role in assigning rooms or laboratories to an in-charge. However, before the lab in-charge could be assigned, registration was necessary, as depicted in Figure 7.

The screenshot shows the 'LABTRACK' application interface for the 'Users' section. The sidebar is the same as in Figure 6. The main content area is titled 'Users' and includes a search bar and a table of user details. The table has columns for Name, Type, Username, and Action. There are five rows of data. An 'Add User' button is in the top right, and a dropdown menu is open for the first row's 'Action' column.

Name	Type	Username	Action
Carl Dawson	Laboratory Incharge	lab2	Action + Edit deactivate Change Password
Custodian	Property Custodian	admin	
Escanor Fauzi	Laboratory Incharge	lab3	
Judylye Novo	Laboratory Incharge	lab0	
Kallo	Laboratory Incharge	lab4	

Figure 7. Lab In-charge Details - Property Custodian



This part had to be operational first. Before assigning rooms, the property custodian had to register the lab in-charge on this page.

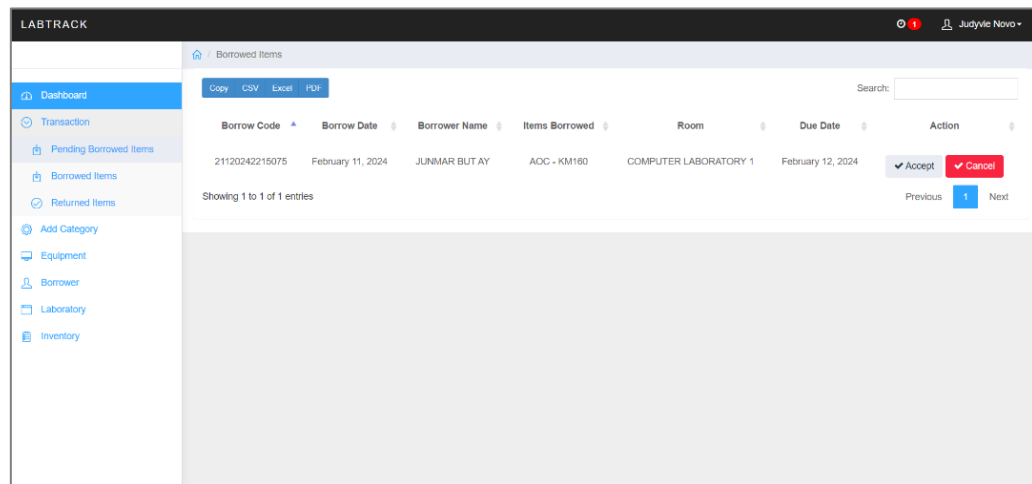


Figure 8. Transactions – Lab In-charge

When a borrower borrowed equipment, a request would appear on this page, under the Transactions button. This was for authorization by the laboratory in-charge to lend the equipment. After accepting the request, the laboratory in-charge would print a receipt form, as depicted in Figure 9.

Category	Item Description
Keyboard	AOC KM160 (SN:SMKM160X0LJ90211056)

Figure 9. Borrowing Receipt Form – Lab In-charge



DON JOSE ECLEO MEMORIAL COLLEGE

P-5 Justiniana Edera, San Jose, Dinagat Islands

Borrow Code	Borrow Date	Borrower Name	Items Borrowed	Room	Due Date	Action
20820241948455	February 08, 2024	JUNMAR BUT AY	DELL - Inspiron 15 3511	COMPUTER LABORATORY 1	February 08, 2024	<button>Return</button>
21120242215075	February 11, 2024	JUNMAR BUT AY	AOC - KM160	COMPUTER LABORATORY 1	February 12, 2024	<button>Return</button>

Figure 10. Borrowed Equipment – Lab In-charge

Once the request receipt was printed, the transaction would be displayed in Figure 10. Additionally, when the borrower returned equipment, a button under transactions labelled Borrowed Items would allow the lab in-charge to verify the returned equipment and make remarks on its condition, whether it was good, lost, or damaged.

Following the return of an equipment, the details would be displayed in the interface shown in Figure 11.

Borrower Name	Returned Items	Borrowed Date	Returned Date	Returned Status
junmar but ay	MICROPACK - M101	February 07,2024	February 07,2024 11:18:39 AM	GOOD
junmar but ay	DELL - Inspiron 15 3511	February 08,2024	February 08,2024 13:50:10 PM	GOOD
junmar but ay	DELL - Inspiron 15 3511	February 08,2024	February 08,2024 13:50:50 PM	GOOD
junmar but ay	MICROPACK - M101	February 08,2024	February 08,2024 13:53:55 PM	DAMAGE

Figure 11. Returned Equipment – Lab In-charge



QR	Model	Category	Brand	Quantity	Quantity Available	Status	Action
	M101	Mouse	MICROPACK	1	0	1 Damage 0 Lost 0 Un-return	More Info
	KM160	Keyboard	AOC	1	0	0 Damage 0 Lost 1 Un-return	More Info
	Inspiron 15 3511	Computer	DELL	1	1	0 Damage 0 Lost 0 Un-return	More Info
	Inspiron 15 3511	Computer	DELL	1	0	0 Damage 0 Lost 1 Un-return	More Info

Figure 12. Equipment – Lab In-Charge

In this interface, the laboratory in-charge could add equipment available for borrowing. They would then generate a QR code for each added equipment.

Date	Item Description	Quantity	Quantity Available	Status
02/07/2024	Mouse M101 MICROPACK	1	0	1 Damage 0 Lost 0 Un-return
02/07/2024	Keyboard KM160 AOC	1	0	0 Damage 0 Lost 1 Un-return
02/07/2024	Computer Inspiron 15 3511 DELL	1	1	0 Damage 0 Lost 0 Un-return
02/07/2024	Computer Inspiron 15 3511 DELL	1	0	0 Damage 0 Lost 1 Un-return

Figure 13. Inventory – Lab In-charge

The lab in-charge could view the inventory of equipment, which included details such as the equipment's condition, whether it was damaged, lost, or unreturned. Additionally, the inventory could be printed based on the equipment's status, month, and year, as shown in Figure 13.

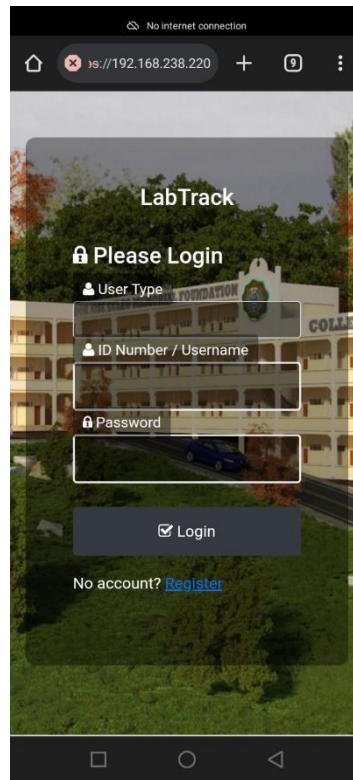


Figure 15. System Authentication (Mobile Devices)

This interface, functions similar to system authentication in desktop mode. Usually this UI for mobile phone is used for borrowers, for their registration and logging in. but it can be also used by the property custodian, and laboratory in-charge.

 Republic of the Philippines DON JOSE ECLEO MEMORIAL COLLEGE San Jose, Dinagat Islands INVENTORY LIST OF ALL ITEM			
Date: February 2024			
Date (M/d/Y)	Item Description	Quantity Available	Item Status
Feb. 07, 2024	Black MICROPACK M101 (S/N: 129200400710)	Purchase Quantity=1 Available Stock=0	1 Damage 0 Lost 0 Un-return
Feb. 07, 2024	Black AOC KM160 (SMKM160D0LJ90211056)	Purchase Quantity=1 Available Stock=0	0 Damage 0 Lost 1 Un-return
Feb. 07, 2024	Gray DELL inspiron 15 3511 (BN9QJ3)	Purchase Quantity=1 Available Stock=1	0 Damage 0 Lost 0 Un-return
Feb. 07, 2024	Gray DELL inspiron 15 3511 (CKJNYJ3)	Purchase Quantity=1 Available Stock=0	0 Damage 0 Lost 1 Un-return
Incharge's Signature			

Figure 14. Printed Inventory

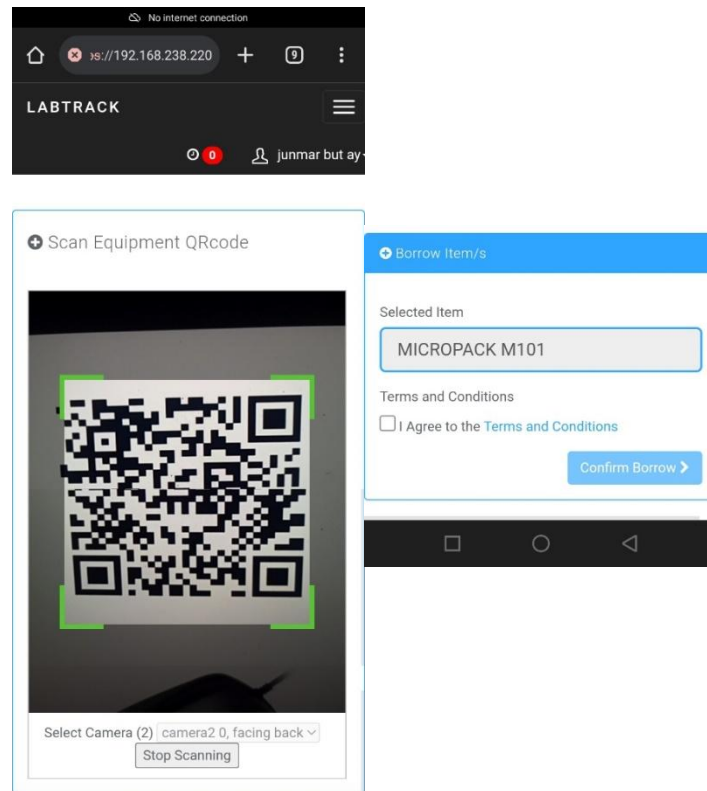


Figure 16. Borrowing Page (QR code scanner)

When the borrower borrowed equipment, this page would be utilized to scan the QR code of the equipment and request agreement regarding the terms and conditions of the equipment before borrowing, in case of returning lost or damaged equipment.

5.2 Evaluation of the System

The International Standard Organization's (ISO) standards were employed for deploying and evaluating the system. Specifically, the system's functionality, efficiency, usability, reliability, maintainability, portability, compatibility, and security were assessed using ISO 25000.



FUNCTIONALITY	MEAN	DESCRIPTION
1. The system has an essential suitable function needed.	3.9	Agree
2. The system has met the guidelines and is appropriately needed by the borrower/property custodian/lab-in-charge.	4	Agree
3. The system has various functions; this system was well integrated and functional.	3.83	Agree
Average	3.91	Agree

Table 3. Functionality of the System

Table 3 illustrated how respondents evaluated the system's performance. The overall mean score was 3.91, indicating a consensus. This suggests that respondents were satisfied with the system's functionality.

Respondents gave the system an average rating for meeting essential functional requirements, having well integrated and functional features, preventing unauthorized access to software functions, adhering to guidelines, being appropriately user-friendly, and providing accurate information. The system and its ability to meet the needs of the responders were deemed satisfactory.

RELIABILITY	MEAN	DESCRIPTION
1. The system is functional.	4	Agree



2. The system can save all the necessary data for the borrowing process.	3.96	Agree
3. The system gives more information necessary to its user.	4.13	Agree
Average	4.03	Agree

Table 4. Reliability of the System

The overall mean score of 4.03 and the consensus across individual questions underscored users' favorable views regarding the system's reliability. The results indicated that the system effectively preserved data, managed errors, facilitated user error recovery, and provided information reliably. Additionally, it maintained the overall functional integrity. The average mean score of 4.03 indicated general agreement regarding the tested dependability characteristics.

USABILITY	MEAN	DESCRIPTION
1. The system is efficient and reliable to use in proving and tracing equipment and borrowers.	4.2	Agree
2. The various functions in the system were well-integrated.	3.86	Agree
3. The system was easy to use.	4.06	Agree
Average	4.04	Agree

Table 5. Usability of the System



With an average mean score of 4.04 across the three usability categories, users indicated their perception that the system was user-friendly. These results imply that users typically viewed the system as effective, intuitive, well-integrated, and conducive to a short learning curve.

EFFICIENCY	MEAN	DESCRIPTION
1. It showcases the execution at the same time the correct output of every process.	3.96	Agree
2. The system is easy to use and user-friendly.	3.93	Agree
3. I can efficiently complete information using this system.	4.03	Agree
Average	3.97	Agree

Table 6. Efficiency of the System

The majority of respondents agreed that the system was efficient. This indicates that the system was well-received in terms of meeting users' needs and expectations. The system in question demonstrated itself as a productive and user-friendly platform, garnering positive feedback across various usability and functionality aspects. The consistent favorable responses validated the system's excellent performance, resulting in an overall mean score indicative of a positive user experience.

MAINTAINABILITY	MEAN	DESCRIPTION
1. The system has the capability of resources to store a huge amount of data.	4.03	Agree



2. The system can store a system/data change	3.96	Agree
3. The system has the capability of speed to execute the process.	3.93	Agree
Average	3.97	Agree

Table 7. Maintainability of the System

The survey's total mean score of 3.97 indicated that respondents considered the system to be well maintainable. They commended the system's code clarity, its adaptability to changes, and the ease of maintenance.

PORTABILITY	MEAN	DESCRIPTION
1. The software can easily replace other software.	4	Agree
2. The software can be accessed/installed easily?	4.16	Agree
3. The software adapts to different environment, and facilities.	4.06	Agree
Average	4.07	Agree

Table 8. Portability of the System

The poll results indicated a generally positive attitude towards the system's mobility, with an overall mean score of 4.07. While most respondents agreed that the approach could be effectively utilized in various circumstances, there were areas



identified for improvement. The findings shed light on crucial aspects of the system's mobility.

COMPATIBILITY	MEAN	DESCRIPTION
1. The system is compatible across different computers and mobile devices.	4.2	Agree
2. The system work well, ensuring the widespread accessibility in any of laboratory.	4.16	Agree
3. The system enhances the process of borrowing using QR code scanner.	4.23	Agree
Average	4.19	Agree

Table 9. Compatibility of the System

The average score of 4.19 across three compatibility factors indicated a consensus regarding the system's compatibility. These findings suggested that, overall, users viewed the system as compatible, easy to use, and well-integrated.

SECURITY	MEAN	DESCRIPTION
1. The Log in process of the system is secure.	3.86	Agree
2. It ensures the privacy, confidentiality, and integrity of data along with the process.	4.1	Agree
3. I am comfortable in providing information about myself that the system requires.	4.26	Agree
Average	4.07	Agree



Table 10. Security of the System

The survey's findings demonstrated a strong evaluation of the system's security measures, achieving an overall mean score of 4.66. Respondents consistently praised the system's capability to safeguard against potential threats and preserve sensitive data. Positive sentiments were observed across various security aspects, as evidenced by the average scores received for specific questions concerning security elements like data encryption, access restrictions, and threat detection.

SYSTEM EVALUATION IF TERMS OF:	MEAN	DESCRIPTION
Functionality	3.91	Agree
Reliability	4.03	Agree
Usability	4.04	Agree
Efficiency	3.97	Agree
Maintainability	3.97	Agree
Portability	4.07	Agree
Compatibility	4.19	Agree
Security	4.07	Agree
Average	4.03	Agree

Table 11. System Evaluation Result



The results of the system evaluation indicated an overall positive response, with an overall mean of 4.03, signifying a strong vote of confidence. The "Agree" ratings reflect users' satisfaction with functionality, compatibility, usability, dependability, security, and maintainability. While efficiency and portability both received "Agree" ratings, there is a slight suggestion for improvement. This comprehensive study underscores the system's robust performance while also pinpointing potential areas for enhancement to ensure ongoing user satisfaction and optimal functionality.



CHAPTER 6

SUMMARY, CONCLUSION, AND RECOMMENDATION

6.1 Summary

The researchers addressed the challenges in every laboratory at Don Jose Edera Ecleo Memorial College by developing a system called LabTrack, which monitored the equipment and its borrowers in every laboratory. The completed system underwent system usability testing to ensure that all functions and details were met.

6.2 Conclusion

In conclusion, the implementation of the 'LabTrack: Property Inventory and Monitoring System using QR Code' has significantly influenced the borrowing operations and processes in every laboratory, as well as their monitoring. LabTrack has the potential to revolutionize operational procedures across all laboratories at Don Jose Ecleo Memorial College. Expected benefits include digitized operations, improved user experiences, data-driven insights, and enhanced reliability.



6.3 Recommendation

Based on the evaluated results, several recommendations were made to enhance the system:

- **Enhanced Equipment Tracking:** RFID technology should have been utilized to improve equipment tracking.
- **Borrowing Purpose Specification:** The purpose of borrowing should have been incorporated to provide additional reporting when items were borrowed.
- **User Interface Enhancement:** The interface could have been improved to a flat design, considering the system was intended for use in both desktop and mobile modes.



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DON JOSE ECLEO MEMORIAL COLLEGE

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APPENDICES



Appendix A

SURVEY QUESTIONNAIRE

**LABTRACK: PROPERTY INVENTORY AND MONITORING SYSTEM
USING QR-CODE**

System Acceptability Scale

Direction: Fill and check your chosen response.

Name: _____ (Optional) **Date:** _____

Occupation:

☐ Staff

☐ Lab In-charge/Property Custodian

☐ Student

1 – Strongly Disagree

4 - Agree

2 – Disagree

5 – Strongly Agree

3 – Neutral/Uncertain

QUESTIONS	1	2	3	4	5
A. FUNCTIONALITY					
The system has an essential suitable function needed.					
2. The system has met the guidelines and is appropriately needed by the borrower/property custodian/lab-in-charge.					
3. The system has various functions; this system was well integrated and functional.					
B. RELIABILITY					
1. The system is functional.					
2. The system can save all the necessary data for the borrowing process.					
3. The system gives more information necessary to its user.					
C. USABILITY					



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1. The system is efficient and reliable to use in proving and tracing equipment and borrowers.					
2. The various functions in the system were well-integrated.					
3. The system was easy to use.					
D. EFFICIENCY					
1. It showcases the execution at the same time the correct output of every process.					
2. The system is easy to use and user-friendly.					
3. I can efficiently complete information using this system.					
E. MAINTAINABILITY					
1. The system has the capability of resources to store a huge amount of data.					
2. The system can store a system/data change					
3. The system has the capability of speed to execute the process.					
F. PORTABILITY					
1. The software can easily replace other software.					
2. The software can be accessed/installed easily?					
3. The software adapts to different environment, and facilities.					
G. COMPATABILITY					
1. The system is compatible across different computers and mobile devices.					
2. The system work well, ensuring the widespread accessibility in any of laboratory.					
3. The system enhances the process of borrowing using QR code scanner.					
H. SECURITY					
1. The Log in process of the system is secure.					
2. It ensures the privacy, confidentiality, and integrity of data along with the process.					
3. I am comfortable in providing information about myself that the system requires.					



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Appendix B

MANUSCRIPT EDITOR CERTIFICATE

CERTIFICATION OF EDITING

This is to certify that the research manuscript titled **"HOME SHARE SEARCH: "LABTRACK: PROPERTY INVENTORY AND MONITORING SYSTEM USING QR-CODE"** authored by **ALEGADO, GENIVIEVE B., BUTAY, JUNMAR D., COQUILLA, JAN REX I., IGCASAMA, JOSE MARI G.**, has undergone a thorough review process and has been found to meet the required standards of grammar, clarity, and academic integrity.

I hereby certify that the research manuscript has been edited and corrected in accordance with the guidelines provided by our institution, ensuring adherence to scholarly standards and publication ethics.

The author's dedication to scholarly excellence and commitment to academic rigor are duly acknowledged. We commend their efforts in producing a well-written and meticulously researched manuscript.

This certification attests that the research manuscript has met the necessary criteria for publication and is deemed suitable for dissemination in academic and scholarly circles.

DELBERT O. SAYSON, LPT

Manuscript Editor

02-12-24



Appendix C

PLAGIARISM CERTIFICATE



Certificate of Plagiarism

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Appendix D

APPROVAL LETTER.

LETTER TO THE SCHOOL PRESIDENT

January 04, 2024

BENGLIN B. ECLEO, MAEd, DPS
School Vice President
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir;

Greetings of Peace and Joy!

We are the 4th Year students in Bachelor of Science in Information Technology (BSIT) at Don Jose Ecleo Memorial College. In our field of specialization, we are tasked to develop a system entitled as a requirement for the Capstone II project.

In this connection, we are asking for your permission to allow us to conduct a study entitled "**LabTrack: Property Inventory and Monitoring System Using QR code**". Rest assured that all solely for educational purposes.

We are hoping for a favorable response regarding this matter.

Thank you and God bless!

Respectfully yours,

The Researchers:

Igcasama, Jose Mari G.
Alegado, Genivieve B.
But-ay, Junmar D.
Coquilla, Jan Rex I.

Noted:

BENGLIN B. ECLEO, MAEd, DPS
School President



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

LETTER TO THE PROGRAM HEAD

January 04, 2024

JHESORLEY M. LAID

BSIT, Program Head

DJEMC

P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir;

Greetings!

We are the 4th years students of Bachelor of Science in Information Technology (BSIT) at Don Jose Ecleo Memorial College. In our field of specialization, we are tasked to develop a system as a requirement for the Capstone project.

In this connection, we are asking for your permission to allow us to conduct a study entitled "**LabTrack: Property Inventory and Monitoring System Using QR code**". Rest assured that all solely for educational purposes.

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Researchers:

Igcasama, Jose Mari G.

Alegado, Genivieve B.

But-ay, Junmar D.

Coquilla, Jan Rex I.

Noted:

JHESORLEY M. LAID

Program Head, BSIT



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

LETTER TO THE ADVISER

January 04, 2024

JUDYVIE E. NOVO

College Instructor

DJEMC

P-5 Justiniana Edera, San Jose, Dinagat Islands

Dear Ma'am;

Greetings of Peace and Joy!

We are the 4th year students of Don Jose Ecleo Memorial College pursuing a degree of Bachelor of Science in Information Technology (BSIT).

We are humbly sent you a letter to sincerely ask for your assistance and expertise to serve as the adviser for our research study entitled **"LabTrack: Property Inventory and Monitoring System Using QR code"**.

Thank you very much for the kind consideration and the researcher are looking forward to working with you.

Respectfully yours,

The Researchers:

Igcasama, Jose Mari G.

Alegado, Genivieve B.

Coquilla, Jan Rex I.

But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO

Thesis Adviser





DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

LETTER TO THE VICE PRESIDENT FOR ACADEMIC AFFAIRS

January 04, 2024

MARICHU M. ESNARDO, Ma.
Vice President for Academic Affairs
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Dear Ma'am;

Greetings of Peace and Joy!

We are the 4th year students of Don Jose Ecleo Memorial College pursuing a degree of Bachelor of Science in Information Technology (BSIT).

In this connection, we are asking for your permission to allow us to conduct a study entitled **"LabTrack: Property Inventory and Monitoring System Using QR code"**. Rest assured that all solely for educational purposes.

Thank you very much for the kind consideration and the researcher are looking forward to working with you.

Respectfully yours,

The Researchers:

Igcasama, Jose Mari G.
Alegado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

MARICHU M. ESNARDO, Ma.
Vice President for Academic Affairs



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

JUDYVIE E. NOVO
COMPUTER LABORATORY IN-CHARGE
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Ma'am,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your laboratory, entitled "**LabTrack: Property Inventory and Monitoring System Using QR code**". We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

Thank you as we anticipate for a response of this request.

Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Approved:

JUDYVIE E. NOVO
Instructor/Computer Laboratory In-Charge



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

CHRISTIAN IAN M. MANTE
P.E. LABORATORY IN-CHARGE
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your laboratory, entitled **“LabTrack: Property Inventory and Monitoring System Using QR code”**. We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

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Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO
INSTRUCTOR

Approved:

CHRISTIAN IAN M. MANTE
P.E. Laboratory In-Charge



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

HERMENIA B. ENSOY, RCrim
CRIME LABORATORY IN-CHARGE
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Ma'am,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your laboratory, entitled "**LabTrack: Property Inventory and Monitoring System Using QR code**". We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

Thank you as we anticipate for a response of this request.

Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO
Instructor

Approved:

HERMENIA B. ENSOY, RCrim
Crime Laboratory In-Charge



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

DELBERT O. SAYSON
SPEECH LABORATORY IN-CHARGE
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your laboratory, entitled **“LabTrack: Property Inventory and Monitoring System Using QR code”**. We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

Thank you as we anticipate for a response of this request.

Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO
Instructor

Approved:

DELBERT O. SAYSON
Speech Laboratory In-Charge



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

JHEY-AR P. BINGOY
PASTRY LABORATORY IN-CHARGE
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your laboratory, entitled **“LabTrack: Property Inventory and Monitoring System Using QR code”**. We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

Thank you as we anticipate for a response of this request.

Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO
Instructor

Approved:

JHEY-AR P. BINGOY
Pastry Laboratory In-Charge



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

ALEX FERNAN
SCHOOL PROPERTY CUSTODIAN
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your organization, entitled “**LabTrack: Property Inventory and Monitoring System Using QR code**”. We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

Thank you as we anticipate for a response of this request.

Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO
Instructor

Approved:

ALEX FERNAN
School Property Custodian



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

APPROVAL LETTER TO THE BENEFICIARY

January 04, 2024

RYAN MARK M. ESNARDO
SCIENCE LABORATORY IN-CHARGE
DJEMC
P-5 Justiniana Edera, San Jose, Dinagat Islands

Sir,

Greetings of peace and joy!

We are fourth-year students at Don Jose Ecleo Memorial College, pursuing a Bachelor of Science degree in Information Technology (BSIT). As part of the requirements for our course IT-401 Capstone 2 Project, we have been tasked with developing a system within our chosen area of specialization.

In connection with this, we kindly request your permission to conduct a system study within your laboratory, entitled "**LabTrack: Property Inventory and Monitoring System Using QR code**". We are very confident that your office can provide us with enough information relevant to the further success of our project. Please rest assured that all data gathered from this project will be utilized solely for educational purposes.

Thank you as we anticipate for a response of this request.

Respectfully yours,

Researchers:

Igcasama, Jose Mari G.
Alegaado, Genivieve B.
Coquilla, Jan Rex I.
But-ay, Junmar D.

Noted:

JUDYVIE E. NOVO
Instructor

Approved:

RYAN MARK M. ESNARDO
Science Laboratory In-Charge



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

Appendix E

MINUTES OF THE FINAL DEFENSE.

_____ Approved

_____ Disapproved

IT THESIS / CAPSTONE PROJECT TITLE: “LabTrack: Property Inventory and Monitoring System Using QR code.”

ADVISER: Judyvie E. Novo

Suggestions / Recommendations	Suggested by:
- Title: LabTrack: Property Inventory and Monitoring System Using QR code. - The main admin is the property custodian. - Property custodian can also monitor the equipment and borrower - Use RAD Base Methodology	RHEA JEAN GAMUTAN-BELSONDRA,
- All laboratory in-charge in the campus can use the system. - Property custodian can add lab in-charge. - Laboratory in charge can view the inventory of every equipment and in borrowing and returning equipment.	JUNRY T. VALENZUELA,
Add notification of returning equipment	NILGEN S. CUATON,
- Use to register their details like name, ID number, program, department then date time they borrow and what they borrow - Add to the system the time to return equipment.	JHESORLEY M. LAID

Noted by:

Signature

Adviser: **Judyvie E. Novo**

Member: **Marichu M. Esnardo, MA**

Member: **Jhesorley M. Laid**

Member: **Nilgen S. Cuaton, MIT**

Member: **Junry T. Valenzuela, MIT**

Chairman: **Rhea Jean Gamutan-Belsondra, MIT**



Appendix F

INTERVIEW QUESTIONS (DATA GATHERING)

Facility/Laboratory-in-charge

1. What is your duty as Facility/Laboratory-in-charge?
2. What is your requirement in borrowing process? Then, what is the data needed for that requirement?
3. What is the manual process of inventory in the facility/laboratory?
4. Are there any problems that you normally encountered in borrowing process? What are those?
5. Who is responsible in the returned damaged equipment?
6. What will happen if the borrowed equipment was lost?

Supply Officer

1. What is your duty as Supply Officer?
 2. What are the facilities/laboratory present in DJEMC?
 3. What is the requirement in borrowing process? Then, what is the data needed for that requirement?
 4. What is the manual process of inventory in DJEMC?
 5. Are there any problems that are normally encountered in borrowing process? What are those?
 6. Who is responsible in the returned damaged equipment?
 7. What will happen if the borrowed equipment was lost?
 8. What do you think/expect of the study entitled “Property Inventory and Monitoring System using QR-based”?
-



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

Appendix G. Certificate of Implementation



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

CURRICULUM VITAE

Name : GENIVIEVE BASALO ALEGADO

Age : 22

Citizenship : Filipino

Sex : Female

Mobile : 09308244295

Email : alegadogenivieve9@gmail.com

Civil Status : Single

Date of Birth : January 26, 2002

Place of Birth : Zone A-1 Pagkakaisa, Sucat, Muntinlupa City

Address : New Mabuhay, Dinagat, Dinagat Islands

Height : 4'7 Ft

Weight : 65 Kg

Religion : Catholic

Father's Name : Geraldine T. Alegado Sr.

Mother's Name : Julieta P. Basalo



EDUCATION

Tertiary Education

DON JOSE ECLEO MEMORIAL COLLEGE
Justiniana Edera, Sanjose, Dinagat Islands

Secondary Education

DINAGAT SCHOOL OF FISHERIES
Dinagat, Dinagat Islands

Primary Education

NEW MABUHAY ELEMENTARY SCHOOL
New Mabuhay, Dinagat, Dinagat Islands



DON JOSE ECLEO MEMORIAL COLLEGE

P-5 Justiniana Edera, San Jose, Dinagat Islands

CURRICULUM VITAE

Name : JUNMAR DELATORRE BUT-AY

Age : 24

Citizenship : Filipino

Sex : Male

Mobile : 09631229982

Email : junmardelatorrebutay@gmail.com

Civil Status : Single

Date of Birth : December 20, 2000

Place of Birth : P-1 Ritaglenda, Basilisa, Dinagat Islands

Address : P-1 Ritaglenda, Basilisa, Dinagat Islands

Height : 5'6 Ft

Weight : 55 Kg

Religion : Filipinista

Father's Name : Ely G. But-Ay

Mother's Name : Mary Jane E. But-Ay



EDUCATION

Tertiary Education

DON JOSE ECLEO MEMORIAL COLLEGE
Justiniana Edera, San Jose, Dinagat Islands

Secondary Education

RITAGGLENDA NATIONAL HIGH SCHOOL
Ritaglenda, Basilisa, Dinagat Islands

Primary Education

RITAGLENDA ELEMENTARY SCHOOL
Ritaglenda, Basilisa, Dinagat Islands



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

CURRICULUM VITAE

Name : JAN REX IMBO COQUILLA
Age : 22
Citizenship : Filipino
Sex : Male
Mobile : 09630731900
Email : janrexcoquilla1@gmail.com
Civil Status : Single
Date of Birth : October 11, 2000
Place of Birth : P-4, Sto.Nino, Libjo, Surigao Del Norte
Address : P-4, Sto.Nino, Libjo, Dinagat Islands
Height : 5'5 Ft
Weight : 50 Kg
Religion : Filipinista
Father's Name : Melchor M. Coquilla
Mother's Name : Teresita I. Coquilla



EDUCATION

Tertiary Education

DON JOSE ECLEO MEMORIAL COLLEGE
Justiniana Edera, San Jose, Dinagat Islands

Secondary Education

ALBOR NATIONAL HIGH SCHOOL
Albor, Libjo, Dinagat Islands

Primary Education

ALBOR CENTRAL ELEMENTARY SCHOOL
Albor, Libjo, Dinagat Islands



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

CURRICULUM VITAE

Name : JOSE MARI G. IGCASAMA
Age : 21
Citizenship : Filipino
Sex : Male
Mobile : 09814223029
Email : josemariigcasama2002@gmail.com
Civil Status : Single
Date of Birth : October 20, 2002
Place of Birth : Alabang, Muntinlupa City, Metro Manila
Address : P-4, Sta. Cruz, Sanjose, Dinagat Islands
Height : 5'8 ft
Weight : 65 kg
Religion : Roman Catholic
Father's Name : Joselito R. Igcasama
Mother's Name : Mariecris G. Igcasama



EDUCATION

Tertiary Education

DON JOSE ECLEO MEMORIAL COLLEGE
Justiniana Edera, Sanjose, Dinagat, Islands

Secondary Education

DON RUBEN EDERA ECLEO SR. MEMORIAL NATIONAL HIGH SCHOOL
Don Ruben, San Jose, Dinagat Island

Primary Education

BURAK ELEMENTARY SCHOOL
Burak, Oras, Eastern Sama



DON JOSE ECLEO MEMORIAL COLLEGE
P-5 Justiniana Edera, San Jose, Dinagat Islands

CURRICULUM VITAE

Name : JUDYVIE E. NOVO
Age : 26
Citizenship : Filipino
Sex : Female
Mobile # : 09485640812
Email : dyviegoodemon@gmail.com
Civil Status : Single
Date of Birth : July 24, 1997
Place of Birth : San Juan, San Jose, Surigao Del Norte
Temporary Address : P-1, San Juan, San Jose, Dinagat Islands
Permanent Address : P-1, San Juan, San Jose, Dinagat Islands
Height : 5'3 Ft
Weight : 55 Kg
Religion : Roman Catholic
Father's Name : Lavvy D. Novo
Mother's Name : Onor E. Novo



EDUCATION

Tertiary Education

DON JOSE ECLEO MEMORIAL FOUNDATION COLLEGE OF SCIENCE AND
TECHNOLOGY
Justiniana Edera, San Jose, Dinagat Islands

Secondary Education

DON RUBEN EDERA ECLEO, SR. MEMORIAL NATIONAL HIGH SCHOOL
DON RUBEN, San Jose, Dinagat Islands



DON JOSE ECLEO MEMORIAL COLLEGE

P-5 Justiniana Edera, San Jose, Dinagat Islands

Primary Education

SAN JUAN ELEMENTARY SCHOOL

Poblacion, San Jose, Dinagat Islands

WORK EXPERIENCE

GIP (2017)

TESDA

Mahayahay, San Jose, Dinagat Islands

ON-THE-JOB TRAINING (2020)

TESDA

Mahayahay, San Jose, Dinagat Islands

ICT FACULTY (PRESENT)

DJEMFCST

Justiniana Edera, San Jose, Dinagat Islands
